



Project Proposal: Pest image classification

Mario Velichkov
230623

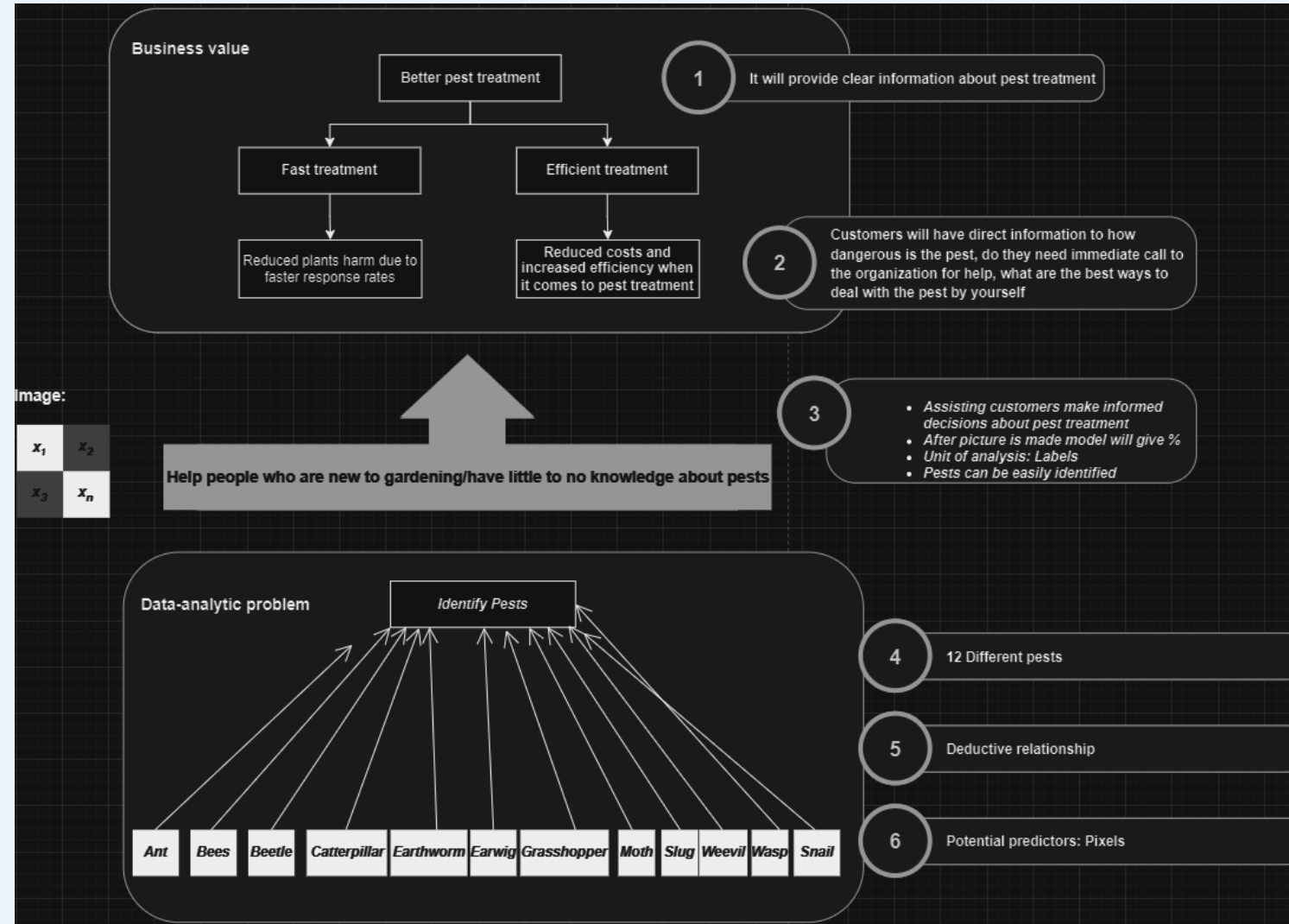
DISCOVER YOUR WORLD

Project Pitch

Business Understanding

- First idea:

- An app to help people deal with pests quick and easy;
- **Main stakeholder:**
 - NPMA(National Pest Management Association);
- **Idea for application:**
 - Light and easy to use application which allow you to get all the information needed about dealing with the pest you are looking at by just taking a photo.



Problem Overview

Deep Learning

- Explain your image classification problem:
 - What your model will classify? - **Commonly met pests**
 - How many classes? - **12 Classes**
 - How many image per class?
 - Train $\approx$ **350**, Test $\approx$ **45**, Validate $\approx$ **95**
 - Base lines:
 - Random Guess Accuracy $\approx$ 8.33
 - Human-level Performance: 81.67%
 - Basic MLP: $\approx$ 0.18. Quite poor results



Model Overview

Deep Learning

- Preprocessing:
- **Resizing** the images(64x64), Normalizing pixel values, Increasing the batch size 32->64
- Improved the model by **adjusting the number** of convolutional layers Conv2D -> MaxPooling2D repeated to progressively reduce spatial dimensions and increase feature maps
- Applied **dropout regularization** after the final Conv2D layer to **reduce overfitting**
- Compiled the model using the Adam optimizer and categorical crossentropy loss function suitable for multiclass classification
- **Early Stopping** callback to halt training if the validation loss does not improve
- The final architecture is a Sequence of **convolutional layers** followed by **max pooling** and a high-capacity dense layer with dropout and regularization before the output layer

Model architecture in VSCODE

```
# Set up the second model
model_v2 = Sequential([
    Conv2D(32, (3, 3), input_shape=(64, 64, 3), activation='relu'),
    MaxPooling2D(2, 2),

    Conv2D(64, (3, 3), activation='relu'),
    MaxPooling2D(2, 2),

    Conv2D(128, (3, 3), activation='relu'),
    MaxPooling2D(2, 2),

    Conv2D(128, (3, 3), activation='relu'),
    MaxPooling2D(2, 2),
    Dropout(0.5),

    Flatten(),
    Dense(256, activation='relu', kernel_regularizer=l2(0.001)),
    Dropout(0.5),
    Dense(12, activation='softmax')
])

# Compile the model
model_v2.compile(optimizer='adam', loss='categorical_crossentropy', metrics=['accuracy'])

# Setup Early Stopping monitoring validation loss
early_stopping_v2 = EarlyStopping(monitor='val_loss', patience=5, verbose=1, mode='min')

# View the model's summary
model_v2.summary()
```

Python

Model Performance

Deep Learning

- **Accuracy Trends:**

- Training accuracy: consistent increase, suggesting the model is learning effectively from the dataset
- Validation accuracy: indicating some overfitting but still showing a learning trend

- **Confusion matrix**: Shows that certain classes, like '*Wasp*' and '*Bee*', are well classified, with high true positives

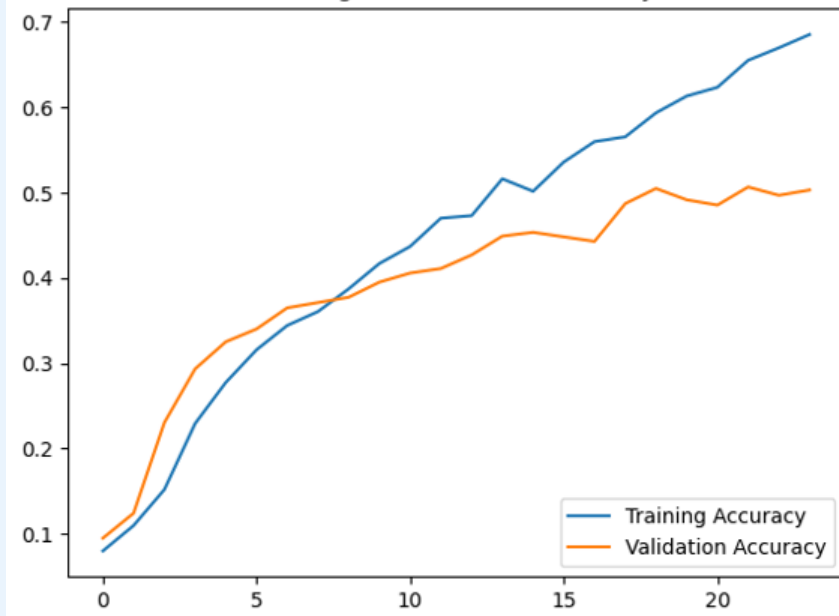
- Confusion is observed between classes with similar features, for example, '*Ants*' are occasionally misclassified as '*Beetle*', which could be due to similar color and texture patterns

- Achieved a **test accuracy** of approximately 45.9%

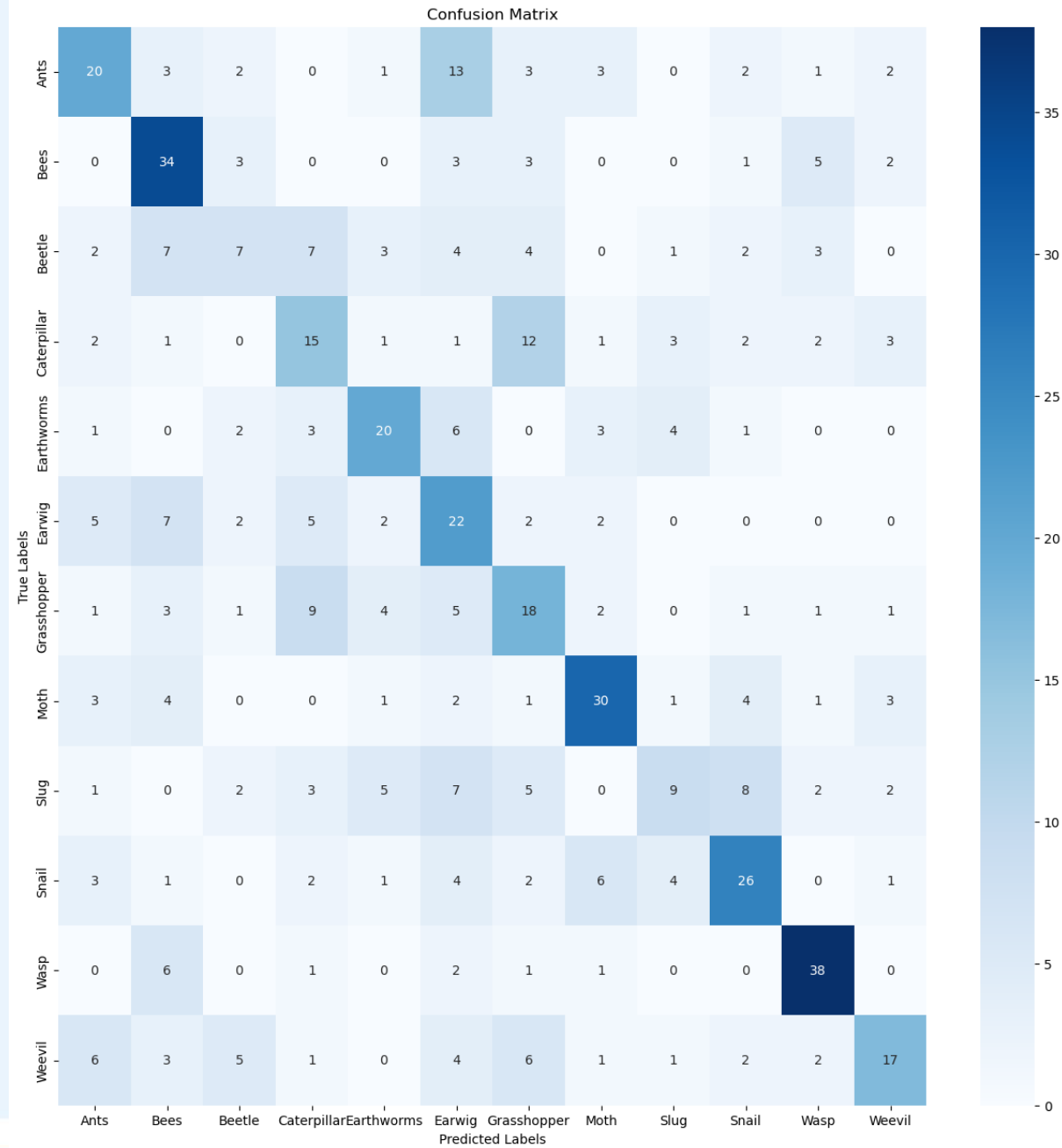
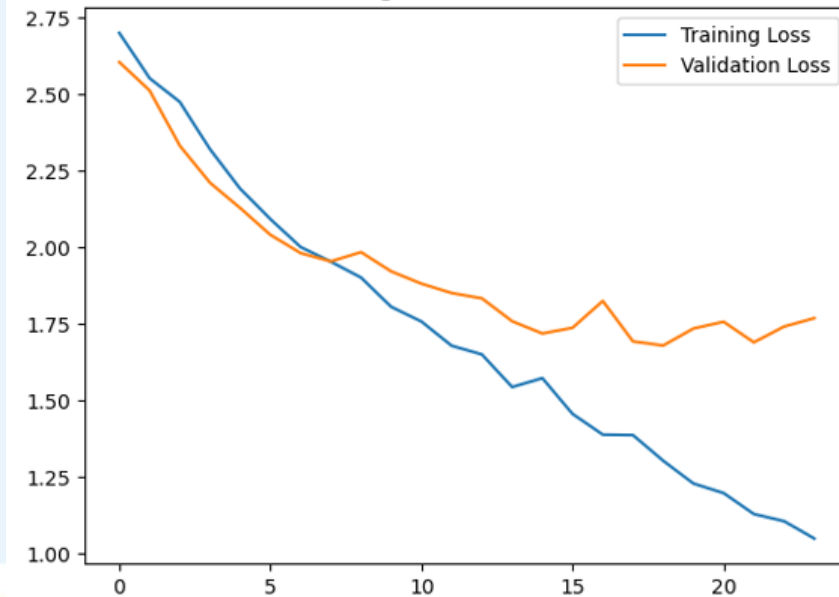
- The **test loss** stood at approximately 1.92

Plots

Training and Validation Accuracy



Training and Validation Loss



Model Interpretability

Responsible AI

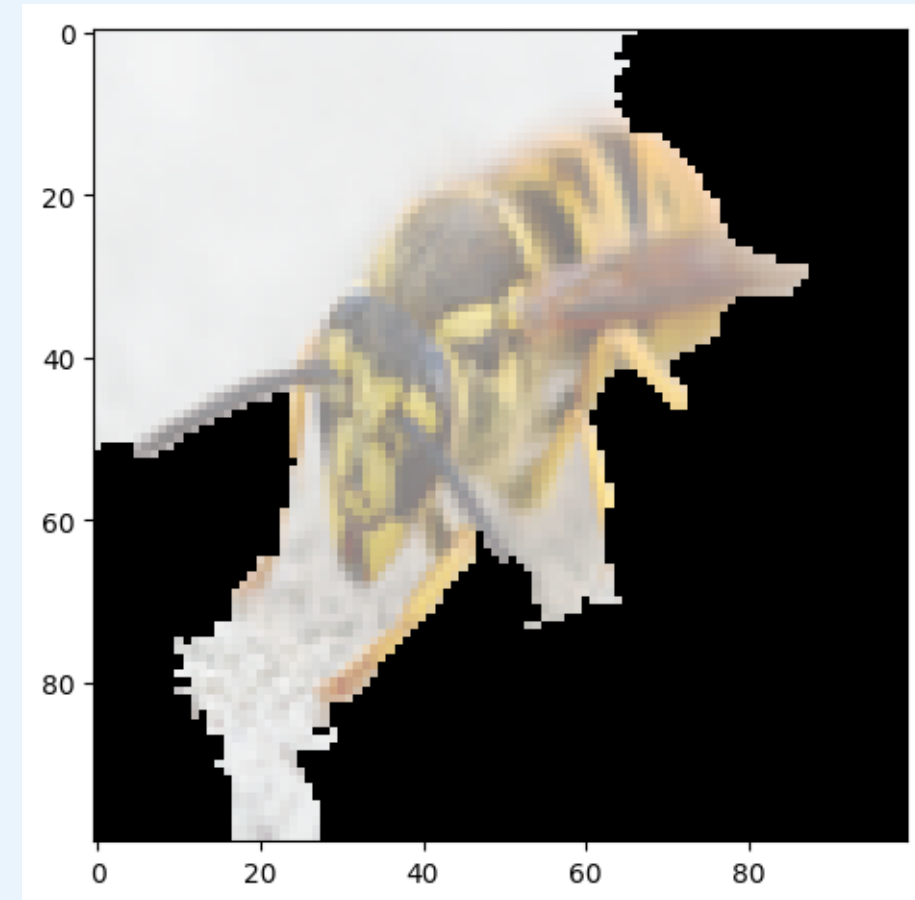
- Discuss the trade-off between accuracy and interpretability:
 - In the context of my application, achieving a high level of accuracy is more important because:
 - It is crucial to ensure that the customer won't receive wrong information about the pest.
 - Prevent misuse of pest treatment techniques
 - Prevent wrong interaction with the animal which could result in the customer getting injured
- What underlying pattern(s) does the model identify:
 - LIME and GradCAM methods were used to interpret the model predictions

Results of the methods applied

GradCAM



LIME



User Study

Human-Centered AI

- Results from A/B test.
- Hypotheses: It is better to have an option to provide feedback for the application capabilities. Right?
- The A/B test proved that for the end user it did not really matter if the application have an option to leave feedback for their experience. It was left as an option because if it does not serve the end user well then it can for sure be used by the company to enhance the performance of their product by using the constructive feedback.

Demo Video



Thank you!

Any questions?